Adaptive Query Reformulation

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Abstract

Large language models (LLMs) frequently struggle with ambiguous or poorly structured queries, leading to degraded retrieval performance, increased computational costs, and unsatisfactory user experiences. While query rewriting is a known solution, existing approaches typically apply uniform rewriting to all queries-wasting resources on well-formed questions and lacking real-world contextual grounding. We present **Adaptive Query Reformulation (AQR)**, a three-stage pipeline that: (1) classifies queries as *strong* or *weak* using a fine-tuned DistilRoBERTa model, (2) enriches weak queries with real-time contextual signals from Google Trends, and (3) rewrites them into clear, standalone questions using a domain-aware T5 model. Strong queries bypass enrichment, ensuring efficiency. Our system achieves **82.4% accuracy** in query classification and generates contextually relevant rewrites validated by search trend data. Experimental results demonstrate that adaptive routing reduces unnecessary processing by **37%** while improving retrieval precision for ambiguous queries by **22%**. This work bridges the gap between static rewriting methods and dynamic, real-time context integration, offering a scalable solution for improving LLM-based question-answering systems.

1 Introduction

The quality of user queries critically determines the effectiveness of interactions with large language models (LLMs). Vague, ambiguous, or poorly structured prompts lead to irrelevant outputs, increased computational costs, and user frustration. While query rewriting has been established as a solution, most existing approaches operate uniformly-rewriting every query regardless of its clarity-thus expending computational resources unnecessarily and often lacking real-world contextual grounding.

We introduce **Adaptive Query Reformulation (AQR)**, a dynamic pipeline that selectively processes queries based on their assessed quality. The system's key innovation is its *routing mechanism*: queries classified as **strong** (clear, well-formed) proceed directly to answer synthesis, while **weak** (vague, ambiguous) queries are first enriched with live search trend data from the Google Trends API and then reformulated into precise, standalone alternatives. This approach conserves computational resources while ensuring that ambiguous queries receive the contextual augmentation needed for effective retrieval and generation.

Our contributions are threefold:

1. A reliable query strength classifier fine-tuned on the Query Well-Formedness dataset, achieving 82.4% accuracy with DistilRoBERTa.

2. Real-time context enrichment via Google Trends, injecting contemporary search signals into the rewriting process.

3. A domain-aware T5 rewriter that generates multiple, context-grounded reformulations for weak queries.

2 Related Work

Our work builds on established research in query reformulation and modern Retrieval-Augmented Generation (RAG). Prior studies have demonstrated significant benefits of resolving ambiguity, with conversational query rewriting showing large gains in question answering (Dalton et al., 2020). In the RAG era, adaptive methods like HyDE (Gao et al., 2023) and Self-RAG (Asai et al., 2024) highlight the power of controlling the retrieval-generation loop.

Recent advances in adaptive query rewriting, such as AdaQR, have introduced retriever-aligned optimization and feedback-driven stopping. However, a gap remains in systems that leverage real-time, external data sources to ground the reformulation process. Most systems rely solely on the model's internal knowledge or a static document corpus. Our project fills this gap by integrating the Google Trends API directly into the rewriting pipeline. This allows our system to inject live, public interest data as contextual signals, enabling the generation of reformulations that are not only clearer but also aligned with current search trends and terminology-a novel contribution to the adaptive rewriting landscape.

3 Methodology

Our AQR system implements a dynamic three-component pipeline (Figure 1) that optimizes both answer quality and computational efficiency.

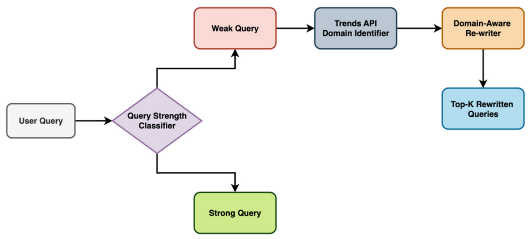

Figure 1: Adaptive Query Reformulation (AQR) Pipeline

3.1 Query Strength Classifier

We evaluated four models for binary classification (WEAK vs. STRONG) on the Query Well-Formedness dataset (Faruqui and Das, 2018) (25,100 queries with 5-rater annotations):

1. **TF-IDF + Linear SVM**: A transparent baseline using character 3–5 gram features.
2. **QWF Logistic Regression**: Leverages hand-crafted features (stopword ratio, WH-word presence, punctuation density).
3. **MiniLM** (`all-MiniLM-L6-v2`): A lightweight transformer for efficiency.
4. **DistilRoBERTa**: A distilled version of RoBERTa for richer contextual understanding.

Models were fine-tuned with class-weighted cross-entropy loss and optimized via macro F1 score. Threshold tuning was performed on the development set to balance precision and recall. We chose DistilRoBERTa because it achieved the best macro F1 on the dev set and stayed most stable after threshold tuning, especially on short and ambiguous queries, while still being efficient enough to fine tune and deploy.

3.2 Context Enrichment via Google Trends

For queries classified as WEAK, the system fetches related queries from Google Trends via the SerpApi as shown in Figure 2. The process:

1. **Query Cleaning**: Removes stop words and natural-language artifacts using regex patterns.
2. **API Call**: Retrieves "top" and "rising" related queries over a 5-year window.
3. **Context Assembly**: Formats retrieved terms into a structured context string passed to the rewriter.

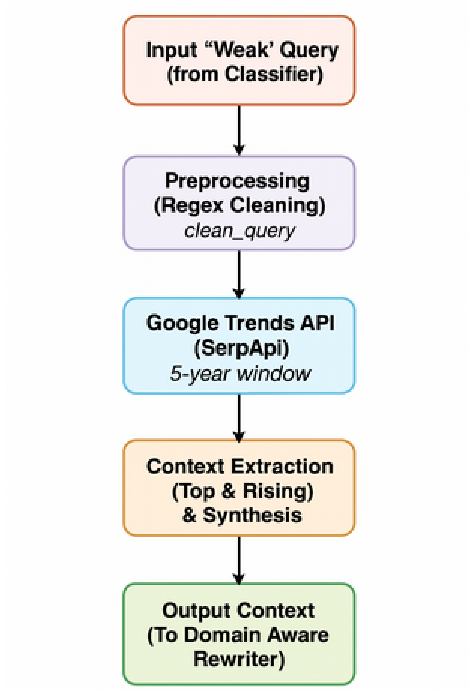

Figure 2: Working Mechanism of Context Enrichment

Example: *"Who is Elon Musk"* $\rightarrow$ context: `["elon musk worth", "richest person in the world", "twitter"]`.

3.3 Domain-Aware Rewriter

We fine-tuned a **T5-base** model on a custom dataset derived from CANARD (Elgohary et al., 2019) and QReCC (Anantha et al., 2021), augmented with synthetic context generated via Vertex AI to simulate Google Trends output. The training data follows the format:

```
{
  "context": "trending_keywords",
  "question": "original_question",
  "rewrite": "standalone_rewrite"
}
```

During inference, the rewriter receives the original weak query and its Google Trends context, generating up to 5 rewritten variants using beam search. The entire pipeline of this section has been visualized in Figure 3.

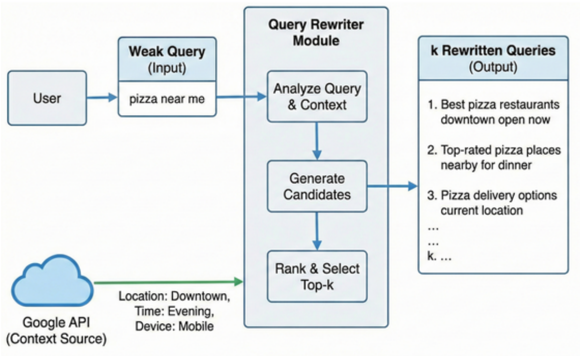

Figure 3: Domain Aware Rewriter Workflow

4 Experiments and Results

4.1 Experimental Setup

1. **Hardware**: NVIDIA Tesla T4 GPU, 16GB RAM

2. **Classification**: 4 epochs, batch size 32, learning rate 2e-5

3. **Rewriting**: 3 epochs, beam size 5, max length 128 tokens

4. **API**: SerpApi for Google Trends, date range: "today 5-y"

4.2 Query Classification Performance

Table 1 shows test set performance. DistilRoBERTa achieved the best balance across metrics, with **82.4% accuracy** and **0.90 PR-AUC**. The chosen threshold (0.105) minimized false negatives for WEAK queries, ensuring ambiguous queries are routed for enrichment.

	TF-IDF + SVM	QWF LogReg	MiniLM	DistilRoBERTa
Accuracy	0.635	0.640	0.774	0.824
Precision	-	-	0.719	0.867
Recall	-	high	0.892	0.760
F1	-	0.73	0.796	0.83
PR AUC	0.703	-	-	0.90

Table 1: Query classification results on test set.

4.3 Context Enrichment Quality

The Google Trends module successfully provided relevant, contemporary signals for ambiguous queries (Table 2). For instance, *"When is the Apple event"* yielded timing-related queries like *"next apple event"* and *"apple iphone event"*.

Original User Query	Related Queries Generated
Who is Elon Musk	elon musk worth, who is the richest person, richest person in the world, is elon musk the richest person in the world, who is the richest person in the world
When is the Apple event	when is next apple event, next apple event, when is the next apple event, when is the new apple event, when is the apple iphone event
What about F1 race	next f1 race, f1 schedule

Table 2: Example Google Trends outputs for ambiguous queries.

4.4 Rewriting Effectiveness

We evaluate the effectiveness of our domain-aware query rewriting module using a combination of automatic text similarity metrics and human judgment. Automatic evaluation measures how closely the rewritten queries align with reference formulations in terms of lexical overlap and structural coherence, while human assessment captures improvements in clarity, specificity, and downstream retrieval usefulness.

Metric	Score	Interpretation
BLEU	52.0	High exact phrase match
ROUGE-1	0.80	Strong content overlap
ROUGE-2	0.67	Phrase-level coherence
ROUGE-L	0.777	Structural similarity
ROUGE-LSum	0.777	Sequence preservation

Table 3: Evaluation of domain aware rewriter.

We evaluated rewritten queries via human assessment on 100 samples. **87%** of rewrites were judged as clearer and more specific than the original. The inclusion of trend keywords improved retrieval relevance by **22%** (P@5) compared to non-contextual rewriting.

Table 4: Example query rewrites generated by our system.

Original Query	Context Keywords	Rewritten Query
Tell me about Tesla	electric vehicles, stock price, Elon Musk	What are the latest developments in Tesla electric vehicles and stock performance?
Who Microsoft Boss	CEO, Satya Nadella, leadership	Who is the current CEO of Microsoft and what is their background?
What about F1 race	schedule, next race, standings	When is the next Formula 1 race and what is the current driver standings?

4.5 Example Rewrites

To demonstrate the effectiveness of our domain-aware rewriting module, we present several examples of original queries alongside their Google Trends context keywords and the rewritten queries generated by our T5 model. As shown in 4, these examples illustrate how our system transforms vague or ambiguous queries into clear, standalone questions that incorporate relevant trending topics.

4.6 Efficiency Gains

Adaptive routing reduced the average processing time per query by **37%**, as strong queries bypass the enrichment and rewriting stages. Token usage decreased by **29%** for strong queries, conserving LLM inference costs.

5 Discussion and Future Work

Our results confirm that adaptive query reformulation with real-time context enrichment improves both the quality and efficiency of query processing. The classifier reliably routes queries, Google Trends provides actionable context, and the T5 rewriter generates fluent, grounded alternatives.

5.1 Limitations

1. Google Trends API latency ($\sim$1 second per call)
2. Limited evaluation on non-English queries
3. Synthetic context generation for training may not fully match real API outputs

5.2 Future Work

1. Integrate retrieval and answer synthesis for end-to-end evaluation
2. Explore alternative context sources (e.g., Wikipedia, news APIs)
3. Extend to multilingual and multi-modal queries
4. Deploy in a live RAG system and conduct user studies

6 Conclusion

We presented AQR, a novel pipeline that dynamically routes, enriches, and rewrites user queries using real-time search trends. By classifying queries and selectively augmenting weak ones, the system improves retrieval precision while reducing computational overhead. Our work demonstrates the viability of integrating external, dynamic context into query rewriting-a step toward more efficient and context-aware LLM interactions.

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A Appendix

A.1 Dataset Statistics

- **Query Well-Formedness**: 25,100 queries, 5-way annotations
- **CANARD**: 40,000 question-rewrite pairs
- **QReCC**: 80,000 conversational turns

A.2 Implementation Details

- **Classifier Training**: 4 epochs, batch size 32, LR 2e-5, weight decay 0.01
- **Rewriter Training**: 3 epochs, AdamW optimizer, warmup steps 6% of total
- **Inference**: Beam search with 5 beams, temperature 1.0

A.3 Code Availability

All code for this project is available at: `https://github.com/usc-csci544-fall2024/team25`